

ROHAN GUPTA

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EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

Los Angeles, CA

MS, Computer Science (emphasis AI & Robotics)

May 2025

Advisor: Professor Jesse Thomason, GLAMOR Lab

UNIVERSITY OF SOUTHERN CALIFORNIA

Los Angeles, CA

BS, Computer Science

May 2024

Activities/Extracurriculars:

- Course Producer for Intro to Artificial Intelligence (CSCI 360) and Intro to Operating Systems (CSCI 350)

PUBLICATIONS

[RobotFleet: An Open-Source Framework for Centralized Multi-Robot Task Planning](#). Rohan Gupta, Trevor Asbery, Zain Merchant, Abrar Anwar, and Jesse Thomason. Oral — *Workshop on Scalable and Resilient Multi-Robot Systems at Robotic Science and Systems*, 2025.

[Efficient Evaluation of Multi-Task Robot Policies With Active Experiment Selection](#). Abrar Anwar, Rohan Gupta, Zain Merchant, Sayan Ghosh, Willie Neiswanger, and Jesse Thomason. Poster — *Conference on Robot Learning*, 2025.

[Contrast Sets for Evaluating Language-Guided Robot Policies](#). Abrar Anwar*, Rohan Gupta*, and Jesse Thomason². Poster — *Conference on Robot Learning*, 2024.

[Exploring Strategies for Efficient Real-World VLN Evaluation](#). Abrar Anwar*, Rohan Gupta*, Elle Szabo*, and Jesse Thomason. Oral — *Workshop on Language and Robot Learning (LangRob) at Conference on Robot Learning*, 2023.

RESEARCH EXPERIENCE

USC VITERBI SCHOOL OF ENGINEERING

Los Angeles, CA

Student Research Assistant, GLAMOR Lab

(May 2023 – June 2025)

- Collaborated with Professor Jesse Thomason in the GLAMOR Lab on robotics, computer vision, and NLP research for vision-and-language navigation and sim2real
- Collected, trained and fine-tuned language-conditioned policies (ACT, Cross Modal Attention, Diffusion, etc.) on multimodal robot data for mobile manipulators and navigation robots using reinforcement learning for finetuning and imitation learning

- Learned ROS1 and ROS2 along a variety of robots' libraries (Toyota HSR, Trossen, Stretch, and some others) and contributed heavily to a working codebase for heterogenous multi-robot coordination project
- Worked on top of public repositories like TeleMoMa and LeRobot and fixed bugs in them as well as added ROS1 to ROS2 upgrades for them
- Built a sample-efficient evaluation framework for a physical robot using perplexity and distance-to-goal as success metrics
- Led efforts on the design of heterogeneous multi-robot collaboration project "RobotFleet", leading design and implementation of an LLM-powered modular framework for end-to-end task planning, task allocation, task execution with natural language commands and a custom standard API across multiple robots
- Set up a Toyota HSR in ROS2 and a Locobot in ROS1 for future use in various research projects

RESEARCH INTERESTS

- **Long-horizon reasoning and hierarchical planning:** structure decision-making over extended timescales using compositional goals and skill hierarchies
- **Robot planning and learning:** combine model-based planning with data-driven learning for adaptive robot behavior in uncertain environments
- **Policy evaluation:** design reliable off-policy and uncertainty-aware evaluation methods for learned policies
- **Program Synthesis for robotics:** generate interpretable robot policies and plans from high-level demonstrations and natural language
- **Multi-robot systems and collaboration:** enable scalable coordination and joint learning across multiple autonomous agents

PROFESSIONAL EXPERIENCE

SYNOPSYS

Sunnyvale, CA

Machine Learning Engineer, Design Technology Group

June 2024 – Present

- Working on a variety of ML projects related to optimizing chip design, using libraries and frameworks like JAX, CUDA, and PyTorch and built featurization pipelines for chip design data, such as netlists and QoR data
- Conducted experiments in JAX and CUDA to speed up differentiable optimization for both ML and non-ML based methods with chip placement algorithms used in products
- Researched and implemented various transformer-based, diffusion-based, and other deep learning model architectures and adapted them for models related to chip design, such as a foundation model for netlists
- Used Python and LangChain to develop multiple LLM-based agents that are set to be incorporated in flagship products like Fusion Compiler, RTLA, and ICC2.

USC INVESTMENT OFFICE

Los Angeles, CA

Software Engineering and Data Analyst Intern

June 2022 – Dec 2022

- Worked with an 8-person team to help run an \$8+ billion endowment by focusing on software and automation projects.
- Made website and coded a financial news webscraper using Python and BeautifulSoup. Performed statistical analysis of financial time-series data using Excel, Python, and other software.

AWARDS AND HONORS

Presidential Merit Scholar (Half-off tuition), **University of Southern California 2020-2024**

Provost's Research Fellowship (Semester Stipend), **University of Southern California Spring 2024**

REVIEWER

Eval & Deploy: Workshop on Evaluation and Deployment in the Robot Learning Lifecycle CoRL 2025

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Java, SQL

Frameworks/Libraries: JAX, CUDA, LangChain, PyTorch, ROS, NumPy, Vector Stores (Milvus, FAISS, etc.), Linux

Tools: WandB, MLFlow, Docker, Git

Software: Foxglove, RViz, Gazebo